

Backend Architecture & Integration Plan

Single backend powering 8 Etsy products + multi-platform marketing (Meta/Google/TikTok) + cross-platform analytics + AI content engine.

Status: 📄 In Planning | **Phases:** MVP → Pro → Full

🌀 **One backend, every channel** — Etsy is the storefront; Meta/Google/TikTok are the funnels; this is the brain.

🔒 **Privacy-first architecture** — server-side tracking via CAPI matches our spreadsheet products' positioning.

Tech Stack

Layer	Choice	Reason
Frontend	Next.js 15 App Router + TypeScript	Already scaffolded
Database	Supabase Postgres + RLS	Already configured
Auth	Supabase Auth + middleware	Already protecting /admin
Storage	Supabase Storage	Files, assets, brand kit
Deploy	Vercel	Auto-deploy from main
Background jobs	Vercel Cron + Supabase Edge Functions	Sync + webhooks
Email	Resend (transactional) + Klaviyo (marketing)	Receipts + campaigns
Tracking	Custom CAPI/SAPI endpoints	Meta CAPI, GA4, TikTok Events

12 Core Backend Modules

- 1. 🛒 Product Catalog**

CRUD admin UI, file uploads per tier, Etsy sync, stats aggregation. Builds on existing products table.
- 2. 📦 Order Intelligence**

Etsy receipt webhook → customer record → fulfillment email (Resend) → signed-URL file delivery (7-day expiry) → log + conversion event.
- 3. 👤 Customer Database (CRM)**

etsy_buyer_id, email, country, total_spend, products_owned, tags, consent flags. Sync to Klaviyo for segmentation.
- 4. 📊 Analytics Hub**

Daily aggregation from Etsy Stats + GA4 + Meta Insights + Google Ads + TikTok Ads + Search Console. Cross-channel ROAS dashboard.

5. 🎯 Ad Campaign Manager

Per-platform campaigns/ad sets/creatives. Pause/duplicate via API. Daily metrics. Banana skill for creative generation.

6. 📡 Pixel & Tracking Manager

Server-side tracking (Meta CAPI, GA4 Measurement Protocol, TikTok Events API). Endpoints: /api/track/page-view, /etsy-click, /lead, /email-signup. iOS 14+ ATT compliant.

7. 📄 Content Publishing Engine

One content atom → many platform renditions (IG, TikTok, FB, Pinterest, X, LinkedIn, Threads, Reddit, YouTube Community, Quora). Review queue + scheduled publishing.

8. 📧 Email Marketing

Klaviyo flows: post-purchase Day 0/3/7/14, weekly digest, win-back, segment-specific drips (Muslim Zakat segment, Small Business segment).

9. ★ Reviews & Reputation

Etsy + Google reviews aggregation. Sentiment flag. AI response template generator.

10. 🔍 SEO Manager

DataForSEO keyword tracking, Search Console integration, sitemap + structured data automation, AI title/meta optimization.

11. 🖨️ AI Content Generator

Etsy listings, ad copy per platform, social posts, email subjects, product photoshoot prompts (banana). Claude (text) + Gemini Nano Banana (images).

12. 💛 Affiliate / Partnership Manager

Tracking links per partner, Pinterest content network, blog/influencer tracking, commission ledger.

Platform Integrations

Etsy

Etsy MCP / Open API v3. Listing CRUD, receipt webhooks (real-time), shop stats (daily), reviews (daily). OAuth 2.0 — token refresh needed.

Meta (Facebook + Instagram)

Graph API v22 + Marketing API + **Conversions API (CAPI)** for server-side tracking + Instagram Graph API for posts/reels + Catalog API for IG/FB Shop. System User long-lived token.

Google

GA4 Data API + Measurement Protocol for server-side traffic tracking. Search Console API for SEO. Google Ads API v17. Merchant Center Content API. Google Business Profile API.

TikTok

TikTok Marketing API v1.3 (ads). **TikTok Events API** for server-side pixel. Content Posting API for organic posts. Business Insights API for analytics.

Pinterest (Tier 2)

API v5 for pins. Pinterest Tag API server-side. Ads API.

Other (Tier 3)

X/Twitter API · LinkedIn API · Reddit API · YouTube Data API

Phasing

Phase 1 — MVP (4–6 weeks)

-  DB schema + RLS (done) ·  Auth middleware (done)
- Product catalog admin UI
- File delivery flow (post-order email + signed URL)
- Etsy order webhook + customer DB
- Resend transactional emails
- Public product pages (storefront)
- Sitemap + robots.txt + llms.txt
- GA4 + Meta Pixel server-side tracking

Phase 2 — Pro (6–8 weeks after MVP)

- Klaviyo + post-purchase flow
- Meta/Google/TikTok ad metrics (read-only)
- Reviews aggregation + Search Console
- AI listing copy generator
- Cross-platform analytics dashboard
- Content atoms + 3 platforms (IG, TikTok, Pinterest)

Phase 3 — Full

- Meta/Google/TikTok ad management (write)
- AI ad creative generator (banana)
- Content publishing — all 10 platforms
- Affiliate manager
- Multi-language storefront (Arabic, Spanish, French)

Estimated Monthly Costs

Service	MVP	Pro	Full
Vercel Pro	\$20	\$20	\$20
Supabase Pro	\$25	\$25	\$25
Resend	\$0-10	\$20	\$50
Klaviyo	—	\$20	\$80
Meta Ads (test)	—	\$300	\$1,500
Google Ads (test)	—	\$300	\$1,500
TikTok Ads (test)	—	\$200	\$1,000
Anthropic API	\$30	\$80	\$200
DataForSEO	—	\$30	\$100
Domain + misc	\$5	\$5	\$10
Total	~\$80	~\$1,000	~\$4,500

Ad spend dwarfs everything; tooling is cheap.

Open Decisions

1. Klaviyo vs. Mailchimp vs. ConvertKit (rec: Klaviyo)
2. Multi-language strategy — separate Etsy listings vs. single English
3. TikTok Shop integration — separate evaluation
4. YouTube as a channel — community posts only or full channel